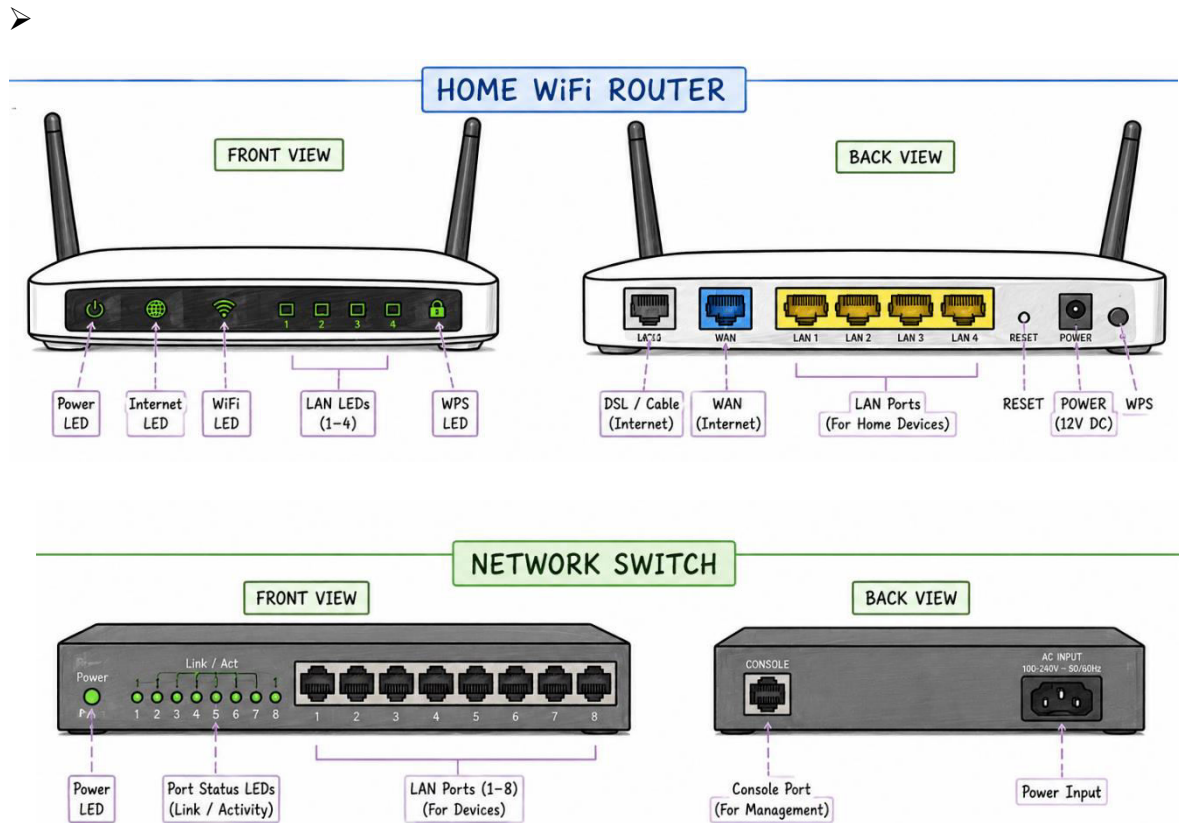


Day 2: Tasks

1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.
Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Feature	Router	Switch
Main Function	A router connects different networks together, such as your home network and the Internet.	A switch connects multiple devices within the same network, such as computers, printers, and access points in an office or home.
How it Handle Traffic	A router uses IP addresses to decide where data should be sent between different networks.	A switch uses MAC addresses to send data directly to the correct device on the same network.

Real-Life Ex.	At home, the router connects your Wi-Fi devices to the Internet provided by your ISP.	In a computer lab or office, a switch connects many computers so they can communicate and share resources
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3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



Router and Switch Identification



Power Port

- Router ke back side par hota hai.
- Isme power adapter connect kiya jata hai.
- Router ko electricity supply deta hai.

Ethernet Port 1 (WAN Port)

- Back side par hota hai.
- Aksar alag color (blue ya yellow) ka hota hai.
- ISP ya modem se internet receive karta hai.

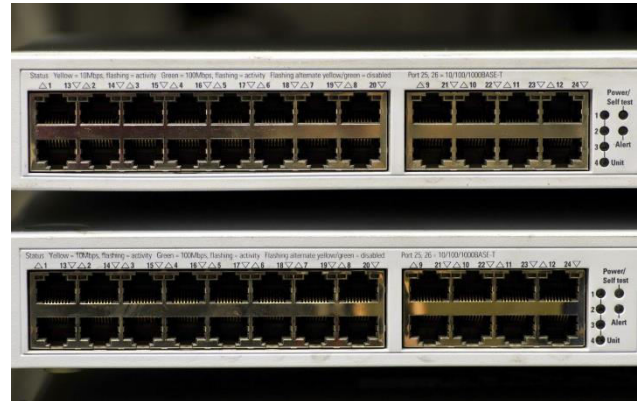
Ethernet Port 2 (LAN Port)

- WAN port ke paas hota hai.
- Computer, laptop ya switch connect karne ke liye use hota hai.

LED Indicators

- Router ke front side par hote hain.
- Power LED router ke ON hone ka status batati hai.
- Internet LED internet connection ka status dikhati hai.
- Wi-Fi LED wireless network active hone ka indication deti hai.

Network Switch



Power Port

- Switch ke side ya back side par hota hai.
- Switch ko power supply dene ke liye use hota hai.

Ethernet Port 1

- Front panel par hota hai.
- Pehla computer ya network device connect karne ke liye.

Ethernet Port 2

- Ethernet Port 1 ke paas hota hai.
- Dusra computer ya network device connect karne ke liye.

LED Indicators

- Har Ethernet port ke paas hote hain.
- Device connected hai ya nahi ye batate hain.
- Data transfer hone par blink karte hain.

Conclusion

- Router internet ko network devices tak pahunchata hai aur Wi-Fi provide karta hai.
- Switch ek hi network ke multiple devices ko connect karta hai.
- Dono devices me Power Port, Ethernet Ports aur LED Indicators hote hain jo power aur network status dikhate hain.

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.



Gaming Café Network Setup

Step 1: Connect the ISP Line

- Sabse pehle ISP (Internet Service Provider) ki internet line ko Router se connect karunga.

- Router internet connection receive karega aur network ko manage karega.

Step 2: Connect the Router to the Switch

- Ek Ethernet (RJ45) cable ki help se Router ko 8-Port Switch se connect karunga.
- Switch network ko multiple devices tak expand karega.

Step 3: Connect the Gaming PCs

- Sabhi 8 Gaming PCs ko alag-alag Ethernet cables se Switch me connect karunga.
- Isse har PC ko internet aur local network access milega.

Step 4: IPL Streaming and Multiplayer Gaming

- Players IPL matches online stream kar sakenge.
- Sabhi PCs ek hi network par hone ke karan multiplayer games smoothly khel sakenge.

Why I Chose a Router First

- Router ISP se internet receive karta hai.
- Router IP addresses assign karta hai.
- Router internet traffic ko manage karta hai.

Why I Used a Switch

- Switch ek saath multiple PCs connect kar sakta hai.
- Wired connection gaming ke liye fast aur stable hota hai.
- Network lag aur delays kam hote hain.

Network Setup Diagram

